Oxygen, kerogen, and morphology: a framework for separating preservational from biological signals across the Great Oxidation Event

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Abstract

The Great Oxidation Event (GOE, ∼2.4 Ga) marks the first persistent rise of atmospheric O_2 above $\sim 10^{-5}$ PAL. Because low oxygen should slow oxidative degradation of organic matter, the anoxic Archean might be expected to preserve organic fossils particularly well; yet morphologically diagnostic microfossils are comparatively rare before the GOE and diversify afterwards. We argue that this paradox dissolves once two preservation modes are distinguished: *bulk organic-carbon survival* and *morphological fidelity.* Redox controls not the total preservation potential alone but the *partitioning* between these modes, via decay pathways and the availability of early-diagenetic mineral templating (silicification, pyritisation, phosphatisation). The GOE-driven rise in seawater sulfate made microbially mediated pyritisation newly available even as rising O_2 sped aerobic decay. We propose a comparative, multiproxy framework—Raman spectroscopy of carbonaceous matter (maturation and structural order), iron speciation, and quantitative morphological-fidelity scoring on lithofacies- and maturity-matched cherts before and after the GOE—integrated through a mechanistic process-graph model and Bayesian data fusion. Four testable predictions can falsify a purely biological reading of the post-GOE fossil expansion and quantify the preservational contribution.

Keywords: Great Oxidation Event; taphonomy; kerogen; Raman spectroscopy; iron speciation; Precambrian microfossils; pyritisation.

1 Introduction: an oxygen–preservation paradox

Before the GOE, atmospheric O_2 was below $\sim 10^{-5}$ PAL—low enough that mass-independent sulfur isotope fractionation (S-MIF), which requires an anoxic atmosphere, was continuously expressed (Pavlov & Kasting, 2002; Farquhar et al., 2000). Archean oceans were dominantly anoxic and ferruginous, with seawater sulfate concentrations <200 µM and perhaps as low as a few µM (Habicht et al., 2002; Crowe et al., 2014). Under such conditions, aerobic respiration—the most efficient organic-matter (OM) degradation pathway—was suppressed, and one might *a priori* expect the Archean to be a high-fidelity archive of early life.

Yet the morphological microfossil record tells a different story. Putatively biological structures are reported from ≥3.4 Ga cherts (Schopf, 1993), but the oldest widely accepted microfossils remain hotly contested (Brasier et al., 2002), and unambiguously diverse, morphologically diagnostic assemblages become abundant only after the GOE—most famously the gunflintia of the ∼1.88 Ga Gunflint Formation (reviewed in Javaux et al., 2018; Wacey et al., 2014). The intuitive expectation—that low oxygen should yield *more* and *better* preserved fossils—is therefore not obviously met.

This paper concerns a single question: *does the post-GOE expansion of the morphological fossil record reflect biological diversification, or a shift in what redox conditions allowed to be preserved?* We do not claim these are mutually exclusive. Instead, we propose a framework to *partition* the observed signal and, where possible, to falsify a purely biological explanation. The central working hypothesis is that *preservation of organic carbon and preservation of morphology are decoupled, and that redox controls their relative availability* (Fig. 1). This reframes the problem from "how much biomass was preserved?" to "what was preservable, and under which regime?"

2 Background

2.1 The GOE and the redox landscape

The GOE—the first persistent rise of atmospheric O_2 above $\sim 10^{-5}$ PAL (Holland, 2002, 2006; Canfield, 2005; Catling & Claire, 2005)—is pinned to ∼2.32–2.45 Ga by the disappearance of mass-independent sulfur isotope fractionation (S-MIF) (Bekker et al., 2004; Farquhar & Wing, 2003), though the transition was prolonged and oscillatory (Lyons et al., 2014; Garduño Ruiz et al., 2024). Proposed drivers include organic-carbon burial during the Lomagundi–Jatuli excursion (Bekker & Holland, 2012; Hodgskiss et al., 2023), hydrogen escape to space (Catling et al., 2001), a shift to subaerial volcanism that weakened the volcanic O_2 sink (Kump & Barley, 2007). Transient pre-GOE "whiffs" of oxygen are reported but contested (Anbar et al., 2007; Gregory et al., 2022). Mid-Proterozoic O_2 remained at ∼1–10% PAL (Lyons et al., 2014) or as low as ≤0.1% PAL (Planavsky et al., 2014), and the deep ocean stayed largely anoxic—ferruginous with euxinic wedges (Canfield, 1998). Critically for what follows, seawater sulfate rose across the GOE from trace (µM) levels to millimolar levels (Habicht et al., 2002; Kah et al., 2004; Crowe et al., 2014). Local redox is reconstructed from iron speciation ($Fe_{HR}/Fe_T > 0.38$; Poulton & Canfield, 2005, 2011), chromium isotopes (Frei et al., 2009), and Mo–U–N systematics (Scott et al., 2008; Garvin et al., 2009). The proxies together document a transition in which the *character* of anoxia changed as much as its spatial extent.

2.2 The Precambrian microfossil record

The Archean microfossil record is sparse and contentious: the filamentous structures of the ∼3.46 Ga Apex chert (Schopf, 1993) were reinterpreted as abiogenic mineral artefacts (Brasier et al., 2002), and even better-preserved Archean microbiotas (Strelley Pool, ∼3.4 Ga; Tumbiana, ∼2.7 Ga) are typically mat-related and morphologically simple. After the GOE the record becomes qualitatively richer, with Gunflint-type assemblages (filaments, coccoids, spindle-shaped forms) widespread in ∼1.9 Ga shallow-marine cherts and taxonomically resolvable (Javaux et al., 2018; Wacey et al., 2012, 2014). The transition from "mats and kerogen" to "diagnostic microfossils" across the GOE is the empirical phenomenon we seek to explain.

2.3 Two modes of preservation

Organic-carbon survival—whether biomass escapes remineralisation and is buried as kerogen—is governed by oxygen-exposure time, sedimentation rate, the intrinsic recalcitrance of biomolecules, and mineral (especially iron-) surface protection (Canfield, 1994; Hedges & Keil, 1995; Hartnett et al., 1998; Lalonde et al., 2012). Anoxia aids OM survival by suppressing aerobic respiration, and ∼20% of sedimentary organic carbon is directly bound to reactive iron phases (Lalonde et al., 2012). *Morphological fidelity*, by contrast, is won or lost in a race between decay and early-diagenetic mineral templating: cellular structure is preserved only when silica, pyrite, phosphate or clay minerals impregnate or encrust cells *before* decay collapses morphology (Wacey et al., 2012). These two modes share decay as a common antagonist but depend on different—and independently variable—mineral pathways. Anoxia therefore favours OM survival unambiguously, but its effect on morphological fidelity is conditional on whether the requisite mineral-templating pathway is available.

2.4 Raman spectroscopy of carbonaceous matter

Raman spectroscopy of carbonaceous material (CM) provides two complementary constraints. Peak-metamorphism thermometry, calibrated by Beyssac et al. (2002), relates the relative area of the defect (D1) and graphite (G) bands—commonly expressed as

$$R_2 = \frac{A_{D1}}{A_{D1} + A_{G1} + A_{G2}} \tag{1}$$

to maximum temperature, allowing samples to be matched for thermal maturity. The D–G parameters and their spatial heterogeneity also report the *structural order* (aromaticity / carbonisation) of the

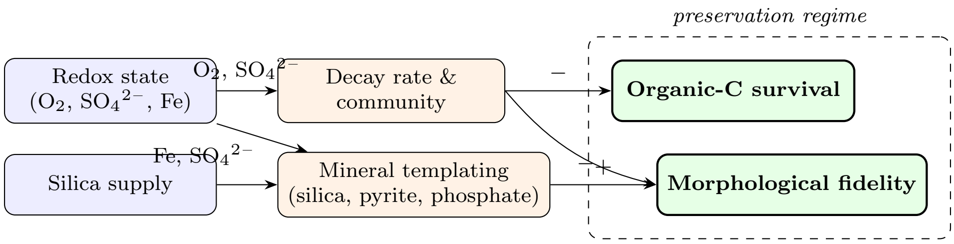

Figure 1: Mechanistic preservation process graph. Redox and silica supply (left) set the decay rate and the availability of mineral-templating pathways (centre), which independently control the two preservation outputs (right). Because mineral-templating availability depends on sulfate and iron, not on O_2 alone, the two outputs can vary independently across the GOE. The graph is the quantitative model fit during data fusion (Sec. 4).

CM (Qu et al., 2019), so that differences in preservation pathway can leave a signature in CM structure beyond the temperature record. Raman mapping thus lets us compare both thermal history and CM structural character across the GOE on a common basis.

3 Hypothesis: a redox-modulated preservation regime

We propose that the GOE did not simply raise or lower "preservation" but shifted the *preservation regime*—the mapping from environmental redox onto the two preservation modes—because the availability of the decisive mineral-templating pathways is itself redox-sensitive in ways that do not track O_2 alone:

- **Pyritisation requires sulfate.** Microbial sulfate reduction yields H_2S that encrusts and replaces cells; this pathway was muted in the sulfate-starved Archean ocean (Habicht et al., 2002; Crowe et al., 2014) and activated only after the GOE (Kah et al., 2004), making a high-fidelity preservation mode newly available even as rising O_2 sped aerobic decay.
- **Silicification depends on the silica cycle, not directly on O_2.** Archean oceans were silica-rich in the absence of biological silica sinkers, so silicification potential was arguably *high* before the GOE—yet Archean silicified assemblages are not correspondingly morphologically rich, suggesting decay commonly outpaced mineralisation, or that mat texture rather than cellular fidelity was preferentially captured.
- **Decay community tracks electron acceptors.** The dominant OM-oxidation pathway shifts with redox (aerobic respiration → sulfate reduction → methanogenesis), changing both the *rate* and the *selectivity* of decay, and hence which morphological characters survive.

This yields a clear, non-trivial expectation: across the GOE, *morphological fidelity* should rise (as pyritisation and other post-GOE pathways activate) even where *bulk OM survival* does not, and possibly where anoxia-driven OM survival declines. The post-GOE "more fossils" signal need not imply more biomass.

4 Proposed framework

4.1 Sample and archival strategy

The central experimental challenge is to isolate redox from confounders—thermal maturity, lithofacies, and rock volume. We therefore propose paired comparison of shallow-marine silicified carbonates and cherts deposited before and after the GOE, matched as closely as possible on lithofacies and on Raman peak maturity (R_2/Tmax), and normalised per unit host rock. Candidate units span the boundary (Table 1). Existing museum collections, drilled cores, and published Raman/kerogen datasets provide immediate material without new fieldwork; a targeted sampling campaign of two or three pre-/post-GOE pairs would anchor new analyses.

Table 1: Candidate units for a lithofacies- and maturity-matched pre/post-GOE comparison. All are shallow-marine, silicified, and low metamorphic grade.

Formation	Age (Ga)	Lithology	Setting	Relative to GOE
Strelley Pool	~3.4	chert/carb.	shallow mar., stromatolitic	pre
Tumbiana Gp.	~2.7	strom. carb./chert	lacustrine–marg. marine	pre
Belingwe (Ngarangan)	~2.7	strom. carb.	shallow marine	pre
Belcher Gp.	~2.0	chert/dolostone	shallow marine	post
Gunflint Fm.	~1.88	chert	shallow mar., foreshore	post
Sokoman IF	~1.88	cherty carb./oxide	shelf	post

4.2 Analytical programme

On each matched sample we propose five measurements:

1. **Raman CM mapping** (R_2 for maturity control; D–G band ratios and spatial heterogeneity for structural order / aromaticity), following the metrology of Beyssac et al. (2002) and the microfossil-application practice of Qu et al. (2019).
2. **Iron speciation** (Poulton & Canfield, 2005) and pyrite iron, to reconstruct local redox and quantify the pyritisation flux.
3. **Total organic carbon** (TOC) and, where possible, CM major-element H/C as an independent maturity/aromaticity index.
4. **Morphological-fidelity scoring** of associated microfossils, performed blind to age and redox, on a defined ordinal scale (cell-wall integrity, internal detail, taxonomic discriminability), with multiple scorers and inter-rater reliability metrics.
5. **Ancillary redox/nutrient proxies** on a subset: sulfur isotopes, Mo–U enrichments (Scott et al., 2008), and δ^{15}N (Garvin et al., 2009).

4.3 Data synthesis, experiments, and integration

We compile published Raman/kerogen, iron-speciation, and microfossil-quality data across the GOE into a unified database, testing—by rarefaction and per-unit-rock normalisation—whether the apparent post-GOE richness survives correction for sampled rock volume. In parallel, laboratory decay–mineralisation experiments on cyanobacterial/mat analogues, factorial in O_2 × sulfate × reactive Fe, would directly measure OM survival against morphological fidelity under each regime, orthogonalising the redox variables that co-varied across the GOE. Finally, the preservation outputs and their drivers (Fig. 1) form a causal graph whose edges carry probabilities and uncertainties: rather than regress "fossil abundance" on environment, we treat the partition as Bayesian inversion over hidden processes (decay regime, pyritisation flux, silicification rate), comparing preservational-shift, biological-expansion, and combined models by marginal likelihood. The graph makes the load-bearing assumption—that decay opposes both outputs while mineral templating supports only morphology—explicit and falsifiable.

4.4 Ruling out alternatives

Four non-redox explanations must be explicitly addressed: (i) *metamorphic overprint*—controlled by matching on R_2/Tmax; (ii) *sampling/rock-volume bias*—addressed by rarefaction and per-volume normalisation; (iii) *true biological expansion*—predicted by the appearance of genuinely novel morphotypes that co-occur with better mineralisation only if expansion is real; and (iv) *facies mismatch*—controlled by lithofacies matching (Table 1).

Table 2: Predictions of the preservational-shift hypothesis and the observations that would distinguish it from a purely biological expansion.

Prediction	Supports preservational shift	Supports biology
P1. Pre-GOE samples, maturity-matched, are OM-rich but morphology-poor, with low pyrite-Fe	high TOC + low fidelity + low Fe_{pyrite}	—
P2. Morphology correlates with mineral-templating proxies, not with TOC/anoxia alone	fidelity $\sim Fe_{pyrite}$ + silica texture	fidelity $\sim$ TOC
P3. The fidelity–OM slope changes across the GOE (interaction term)	significant age×OM interaction	no interaction
P4. In decay experiments, oxic/+sulfate treatments preserve morphology best despite lower OM survival	morphology ↑ while OM ↓	morphology tracks OM

5 Testable predictions

Under a purely biological expansion, none of P1–P3 should hold: morphology would track OM abundance, pyrite would not be predictive, and the fidelity–OM relationship would be invariant across the GOE. The four predictions are thus mutually reinforcing and falsifiable on the proposed data.

6 Implications

If the preservational-shift hypothesis is supported, the post-GOE "rise" of microfossils is partly a taphonomic artefact of redox-driven mineral-pathway activation—most conspicuously sulfate-enabled pyritisation—rather than evidence of a contemporaneous biological radiation. The practical consequences are threefold. First, quantitative estimates of Proterozoic biomass or diversity that treat fossil abundance as a direct biological signal are biased and should be redox-corrected. Second, the timing of genuine biological innovations (e.g. the origin of oxygenic photosynthesis, eukaryotes) must be inferred from preservational proxies, not from first appearances alone. Third, the same logic applies to the search for biosignatures on Mars and in other Archean-analogue settings: the absence of morphological fossils in an anoxic succession does not entail the absence of life, only the absence of the mineral pathway required to record it.

7 Conclusions

We have argued that the apparent paradox of an Archean that is organic-rich yet morphology-poor is resolved by separating two preservation modes that redox affects differently. The GOE's most consequential taphonomic legacy may be the activation of sulfate-driven pyritisation, which preserved morphology even as oxidising conditions nominally hastened decay. We propose a multiproxy, lithofacies- and maturity-matched comparison of pre- and post-GOE cherts—Raman CM, iron speciation, quantitative morphology—embedded in a mechanistic process-graph model with Bayesian data fusion, and we give four falsifiable predictions. The framework is designed so that a purely biological account of the post-GOE fossil expansion can be tested, and—if it fails—so that the preservational contribution can be quantified rather than merely asserted.

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